

Real-Time Semantic Segmentation on FPGA for Autonomous Vehicles Using LMIINet with the CGRA4ML Framework

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Abstract—Semantic segmentation has emerged as a fundamental problem in computer vision, gaining particular importance in real-time applications such as autonomous driving. The main challenge is achieving high accuracy while operating under computational and hardware constraints. In this research, we present an FPGA-based implementation of real-time semantic segmentation leveraging the lightweight LMIINet architecture and the Coarse-Grained Reconfigurable Array for Machine Learning (CGRA4ML) hardware framework. The model was trained using Quantization-Aware Training (QAT) with 8-bit precision on the Cityscapes dataset, reducing memory footprint by a factor of four while enabling efficient fixed-point computations. Necessary modifications were applied to adapt the model to CGRA4ML constraints, including simplifying skip connections, employing hardware-friendly operations such as depthwise-separable and 1A-1 convolutions, and redesigning parts of the Flatten Transformer. Our implementation achieves approximately 90% pixel accuracy and 45% mean Intersection-over-Union (mIoU), operating in real-time at 20 frames per second (FPS) with 50.1 ms latency on the ZCU104 FPGA board. The results demonstrate the potential of CGRA4ML, with its flexibility in mapping modern layers and off-chip memory utilization for skip connections, provides a path for implementing advanced semantic segmentation networks on FPGA for real-time applications to outperform traditional GPU solutions in terms of power efficiency while maintaining competitive accuracy. The code for this project is publicly available at https://github.com/STAmirr/cgra4ml_semantic_segmentation.

Index Terms—FPGA, LMIINet, semantic segmentation, real-time inference, CGRA4ML, autonomous vehicles, hardware acceleration, quantization-aware training (QAT)

I. INTRODUCTION

RELIABLE autonomous driving demands dense scene understanding at video rates under strict power, thermal, and cost constraints. Among perception tasks, *semantic segmentation* provides pixel-wise interpretation of a scene—separating roadway, sidewalk, vehicles, pedestrians, and signage—and directly supports planning and control. Over the last decade, accuracy has improved rapidly with deeper backbones and attention/Transformer modules; however, typical GPU deployments remain power-hungry and introduce non-deterministic latency due to driver scheduling, memory traffic, and batching requirements. These properties are problematic for embedded electronic control units (ECUs) that

must meet hard end-to-end deadlines and functional-safety constraints in automotive environments.

A. Why FPGA for real-time semantic segmentation.

Field-programmable gate arrays (FPGAs) provide fine-grained spatial parallelism and deterministic, clocked pipelines with tight control over the memory hierarchy. Compared to general-purpose GPUs, the accelerator datapath can be specialized to the exact operations in the network (e.g., depth-wise separable convolutions, pointwise convolutions, attention projections), enabling:

- **Deterministic latency:** deep pipelines with static initiation intervals (II) and bounded queues, avoiding OS and driver jitter. End-to-end latency scales with pipeline depth rather than input batch size; real-time inference at batch $b=1$ is natural rather than penalized.
- **Reduced data movement:** line buffers and on-chip SRAM tile the feature maps to minimize off-chip DRAM traffic—a dominant source of latency and energy on GPUs. Streaming between layers eliminates intermediate host-device transfers.
- **Quantization-first arithmetic:** fixed-point INT8/INT4 MAC arrays match the needs of quantization-aware training (QAT), lowering energy per operation while preserving accuracy under careful calibration.
- **Operator fusion and dataflow execution:** custom pipelines fuse convolution, normalization, and activation, collapsing kernel boundaries and removing launch overheads.
- **Safety and lifecycle:** reconfigurability enables over-the-air updates and field patches while preserving a verified timing budget—useful for ISO 26262 processes.

B. From FPGA to ASIC: The Migration Path and Cost Rationale

A hardware-software co-design validated on FPGA establishes the micro-architecture (compute arrays, buffer sizes, dataflow, and quantization) that can later be hardened as an ASIC. While ASICs incur non-recurring engineering (NRE) costs, at production scale their unit cost and energy per inference are substantially lower than GPU modules or large FPGAs. Crucially, the design techniques we employ—tiling sizes, reuse factors, loop unrolling, operand bit-widths, and on-chip buffer topology—transfer directly to an ASIC netlist. Thus, the same design that yields deterministic ≤ 50 ms

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